

ETC2410 Assignment2

Group 88

Question 1

(a)

$$\widehat{\text{price}} = \underset{(29.280)}{-21.660} + \underset{(0.001)}{0.002} \cdot \text{lotsize} + \underset{(0.013)}{0.121} \cdot \text{sqrft} + \underset{(9.241)}{9.590} \cdot \text{bdrms} + \underset{(14.490)}{20.070} \cdot \text{colonial} \quad (1)$$

(b)

Definition of Homoskedasticity

In the context of this regression model, homoskedasticity means that the variance of the error term u_i is constant across all observations. Formally:

$$\text{Var}(u_i | \text{lotsize}_i, \text{sqrft}_i, \text{bdrms}_i, \text{colonial}_i) = \sigma^2, \quad \forall i$$

That is, the variability of the unobserved factors affecting house price does not depend on lot size, square footage, number of bedrooms, or whether the home is colonial style.

Definition of Heteroskedasticity

Conversely, heteroskedasticity occurs when the variance of the error term u_i is not constant but instead depends on one or more explanatory variables. Formally:

$$\text{Var}(u_i | \text{lotsize}_i, \text{sqrft}_i, \text{bdrms}_i, \text{colonial}_i) = \sigma_i^2, \text{ where } \sigma_i \text{ varies with } i$$

In this context, it means that the spread of house prices unexplained by the model could differ systematically with, for example, house size or neighborhood characteristics — larger or more expensive houses might have more variable pricing errors than smaller ones.

(c)

We expect the error term in equation (1) to be **heteroskedastic**.

Reason: House prices vary widely, and unobserved factors such as location, building quality, and renovation status tend to cause larger prediction errors for more expensive or larger houses. Given the *cross-sectional* nature of the data and that predictors such as `sqrft` and `lotsize` vary substantially in scale, the assumption of constant variance is implausible.

(d)

Estimated regression

$$price_i = \beta_0 + \beta_1 \cdot lotsize_i + \beta_2 \cdot sqrft_i + \beta_3 \cdot bdrms_i + \beta_4 \cdot colonial_i + u_i$$

The Breusch–Pagan (BP) test evaluates whether the variance of the error term depends on the explanatory variables.

Auxiliary Regression

$$u_i^2 = \delta_0 + \delta_1 \cdot lotsize_i + \delta_2 \cdot sqrft_i + \delta_3 \cdot bdrms_i + \delta_4 \cdot colonial_i + \varepsilon_i$$

Hypotheses :

$$H_0 : \delta_1 = \delta_2 = \delta_3 = \delta_4 = 0$$

$$H_1 : \delta_j \neq 0 \quad \text{for at least one } j \in \{1, 2, 3, 4\}$$

Test Statistic and Distribution:

$$LM = n \times R_{\text{aux}}^2 \sim \chi_4^2$$

Decision Rule

At the 5% significance level critical value: $\chi_{0.95,4}^2 = 9.488$

Decision rule: If $LM > 9.488$ or $p < 0.05$, reject H_0 ; Otherwise, fail to reject H_0 .

Sample Result and Conclusion

From the R output:

$LM = 11.36$, $df = 4$, $p = 0.0228$.

Conclusion: Since $11.36 > 9.488$ and $p = 0.0228 < 0.05$, we reject the null hypothesis of homoskedasticity. There is statistically significant evidence that the conditional variance of the error term varies across houses. Therefore, the error term in the model is **heteroskedastic**. While OLS estimates of coefficients remain unbiased, the standard errors are inefficient and invalid for inference.

(e)

Estimated regression

$$price_i = \beta_0 + \beta_1 \cdot lotsize_i + \beta_2 \cdot qrft_i + \beta_3 \cdot bdrms_i + \beta_4 \cdot colonial_i + u_i$$

Auxiliary Regression

$$\begin{aligned} \hat{u}_i^2 = & \alpha_0 + \alpha_1 lotsize_i + \alpha_2 sqrft_i + \alpha_3 bdrms_i + \alpha_4 colonial_i \\ & + \gamma_1 (lotsize_i^2) + \gamma_2 (sqrft_i^2) + \gamma_3 (bdrms_i^2) \\ & + \tau_{12} (lotsize_i \cdot sqrft_i) + \tau_{13} (lotsize_i \cdot bdrms_i) + \tau_{14} (lotsize_i \cdot colonial_i) \\ & + \tau_{23} (sqrft_i \cdot bdrms_i) + \tau_{24} (sqrft_i \cdot colonial_i) + \tau_{34} (bdrms_i \cdot colonial_i) + v_i \end{aligned}$$

We omit $colonial_i^2$ because $colonial_i$ is a dummy variable, $colonial_i^2 = colonial_i$ hence $k=13$

Hypotheses :

$$\begin{aligned} H_0 : & \text{Var}(u_i | X_i) = \sigma^2 \quad (\text{errors are homoskedastic}) \\ \Rightarrow & \alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = \gamma_1 = \gamma_2 = \gamma_3 = \tau_{12} = \tau_{13} = \tau_{14} = \tau_{23} = \tau_{24} = \tau_{34} = 0 \\ H_1 : & \text{Var}(u_i | X_i) = \sigma_i^2, \text{ and at least one coefficient } \neq 0 \end{aligned}$$

Test Statistic and Distribution:

$$LM = n \times R_{aux}^2 \sim \chi_k^2 = \chi_{(13)}^2$$

Decision Rule

At the 5% significance level:

Critical value:

$$\chi_{0.95,13}^2 = 22.362$$

Decision rule: If $LM > \chi_{0.95,13}^2$ or $p < 0.05$, reject H_0 ; Otherwise, fail to reject H_0 .

Sample Result and Conclusion

From the R output:

$$LM = 30.785, df = 13, p = 0.0036.$$

Since $30.785 > 22.362$ and $p = 0.0036 < 0.05$, we reject the null hypothesis of homoskedasticity.

There is **strong evidence of heteroskedasticity** in the regression residuals. This means the error variance is not constant across observations, violating one of the classical OLS assumptions.

(f)

Method.

Because the White test in part (e) rejected the null of homoskedasticity, the model is estimated again using **heteroskedasticity-robust (HC1)** standard errors. Coefficient estimates remain the same, but the estimated variances and t-statistics change.

Table 1: Q1(f): OLS vs HC1 (heteroskedasticity-robust) standard errors

term	beta	se_OLS	p_OLS	se_HC1	p_HC1
(Intercept)	-21.658	29.276	0.462	37.341	0.563
lotsize	0.002	0.001	0.001	0.001	0.071
sqrft	0.121	0.013	0.000	0.017	0.000
bdrms	9.590	9.241	0.302	8.834	0.281
colonial	20.075	14.494	0.170	15.937	0.211

Interpretation.

The coefficient estimates are unchanged, but several HC1 standard errors are larger than their OLS counterparts, especially for *lotsize* and *colonial*, implying that the residual variance increases with these variables. As a result, the associated ttt-statistics decline in magnitude:

sqrft remains highly significant ($p < 0.01$); *lotsize* falls from strongly significant ($p = 0.001$) to marginally significant ($p = 0.071$); *bdrms* and *colonial* remain insignificant under both.

Conclusion.

Since heteroskedasticity is present, inference should rely on heteroskedasticity-robust (HC1) standard errors rather than the usual OLS ones. This adjustment ensures that ttt-tests and confidence intervals remain valid even when the error variance is not constant.

(g)

Model to Estimate

$$\log(\text{price}_i) = \beta_0 + \beta_1 \log(\text{lotsize}_i) + \beta_2 \log(\text{sqrft}_i) + \beta_3 \text{bdrms}_i + \beta_4 \text{colonial}_i + u_i$$

Estimated Model

$$\widehat{\log(\text{price})} = \underset{(0.317)}{-1.242} + \underset{(0.045)}{0.179} \cdot \log(\text{lotsize}) + \underset{(0.081)}{0.678} \cdot \log(\text{sqrft}) + \underset{(0.028)}{0.029} \cdot \text{bdrms} + \underset{(0.043)}{0.054} \cdot \text{colonial} \quad (2)$$

Interpretation of coefficients

$\log(\text{lotsize})$: The estimated elasticity is 0.179 ($p < 0.001$). Holding other factors constant, a 1 % increase in lot size is associated with a 0.18 % increase in house price.

$\log(\text{sqrft})$: The estimated elasticity is 0.678 ($p < 0.001$). A 1 % increase in floor area leads to a 0.68 % rise in price. The elasticity for floor area is substantially larger than that for lot size, suggesting that interior living space contributes more to property value than land area.

bdrms : The coefficient 0.029 ($p = 0.31$) indicates that each additional bedroom raises price by about 2.9 %, but the effect is not statistically significant.

colonial : The coefficient 0.054 ($p = 0.22$) suggests that colonial-style houses are roughly 5.6 % more expensive than non-colonial homes ($(e^{0.0543} - 1) \times 100\%$), though this effect is statistically insignificant.

Conclusion:

Both *lotsize* and *floor area* have positive effects on price, but only *sqrft* shows a strong, statistically significant elasticity. The number of bedrooms and house style have no significant impact on house prices.

(h)

Estimated regression

$$\log(\text{price}_i) = \beta_0 + \beta_1 \log(\text{lotsize}_i) + \beta_2 \log(\text{sqrft}_i) + \beta_3 \text{bdrms}_i + \beta_4 \text{colonial}_i + u_i$$

Auxiliary Regression

$$\hat{u}_i^2 = \gamma_0 + \gamma_1 \log(\text{lotsize}_i) + \gamma_2 \log(\text{sqrft}_i) + \gamma_3 \text{bdrms}_i + \gamma_4 \text{colonial}_i + \epsilon_i$$

Hypothesis:

Null Hypothesis: $H_0 : \gamma_1 = \gamma_2 = \gamma_3 = \gamma_4 = 0$ (Homoskedasticity)

Alternative Hypothesis: $H_1 : \text{At least one } \gamma_j \neq 0 \text{ for } j = 1, 2, 3, 4$ (Heteroskedasticity)

Test Statistic and Distribution:

$$LM = n \times R_{aux}^2 \sim \chi_{(4)}^2$$

Critical value:

$$\chi_{0.95,4}^2 = 9.488$$

Decision rule: If $LM > \chi_{0.95,4}^2$ or $p < 0.05$, reject H_0 ; Otherwise, fail to reject H_0

Sample Result and Conclusion

From the R output:

$$LM = 6.6063, df = 4, p = 0.1582$$

Conclusion:

Since $6.6063 < 9.4877$ and $p = 0.1582 > 0.05$ do not reject H_0 . There is no evidence of heteroskedasticity at the 5% level. There is no evidence of heteroskedasticity in the log-linear model (2) at the 5% level.

Question 2

(a)

Model to estimate:

$$\log(\text{avgprc}_t) = \beta_0 + \beta_1 \text{mon}_t + \beta_2 \text{tues}_t + \beta_3 \text{wed}_t + \beta_4 \text{thurs}_t + \beta_5 t + \beta_6 \text{wave2}_t + \beta_7 \text{wave3}_t + u_t$$

Estimated equation:

$$\begin{aligned} \widehat{\log(\text{avgprc})} = & \underset{(0.189)}{-0.954} - \underset{(0.114)}{0.012} \cdot \text{mon} - \underset{(0.112)}{0.014} \cdot \text{tues} + \underset{(0.112)}{0.043} \cdot \text{wed} + \underset{(0.111)}{0.122} \cdot \text{thurs} \\ & - \underset{(0.001)}{0.001} \cdot t + \underset{(0.022)}{0.095} \cdot \text{wave2} + \underset{(0.021)}{0.049} \cdot \text{wave3} \end{aligned} \quad (3)$$

Significance of wave variables (individual tests, 5% level):

If the p-value is less than the 0.05 significance level, variable is statistically significant at the 5% level.

$$t_j = \frac{\hat{\beta}_j - 0}{\text{SE}(\hat{\beta}_j)}.$$

wave2 : $\hat{\beta}_6 = 0.0946$, s.e. = 0.0218, $t = 4.342$, $p = 3.73 \times 10^{-5} \Rightarrow$ significant at 1%

wave3 : $\hat{\beta}_7 = 0.0490$, s.e. = 0.0210, $t = 2.331$, $p = 0.022 \Rightarrow$ significant at 5%

Wave2 and wave3 are both significant at 5% level, indicating that the wave periods are significantly affect the price of the fish.

Economic interpretation :

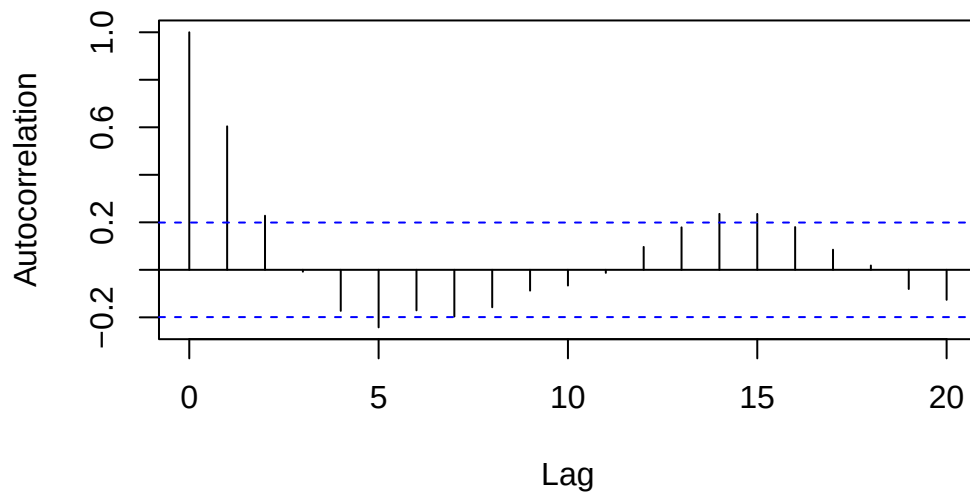
- wave2 $\uparrow$ 1unit $\rightarrow$ price $\uparrow$ about $100 [e^{0.0946} - 1] \approx 9.9\%$.
- wave3 $\uparrow$ 1unit $\rightarrow$ price $\uparrow$ about $100 [e^{0.0490} - 1] \approx 5.0\%$.

Mechanism (why stormy seas raise prices):

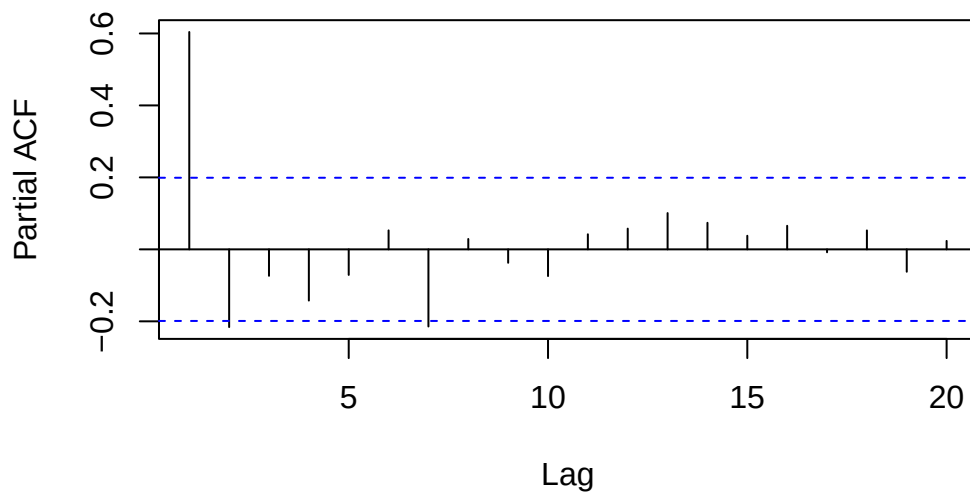
Stormy weather (as measured by wave2 and wave3) increases the risks and costs associated fishing (fuel, time, safety, spoilage), so fewer boats go out and catches fall or are delayed. These supply-side frictions shift the short-run supply curve left, reducing effective market supply. With day-to-day demand roughly unchanged (and relatively inelastic), the equilibrium price rises, which is exactly what the positive and significant wave2/wave3 coefficients indicate: rougher seas in the recent past are associated with higher fish prices.

(b)

Correlogram of OLS Residuals from Model 2(a)



PACF of OLS residuals



Findings from the correlogram :

The ACF of the OLS residuals shows a large, statistically significant spike at lag 1, while correlations at higher lags quickly fall within the 95 % confidence bands. The PACF displays a single pronounced spike at lag 1 and no further significant partial autocorrelations. This pattern indicates that residuals are positively autocorrelated only at the first lag, with no evidence of longer-term or seasonal dependence.

Interpretation and discussion :

The error term therefore exhibits short-run, first-order serial correlation rather than white-noise behaviour. In practical terms, residuals from one day tend to be followed by residuals of similar sign on the next day, implying that part of the short-term dynamics in fish prices is not fully captured by the model.

The correlogram suggests a clear first-order positive autocorrelation in the residuals. Hence, the error term is not white noise, and inference based on conventional OLS standard errors may be unreliable.

(c)

Model under test:

$$\log(\text{avgprc}_t) = \beta_0 + \beta_1 \text{mon}_t + \beta_2 \text{tues}_t + \beta_3 \text{wed}_t + \beta_4 \text{thurs}_t + \beta_5 t + \beta_6 \text{wave2}_t + \beta_7 \text{wave3}_t + u_t$$

Hypotheses :

Let the error follow $u_t = \rho_1 u_{t-1} + \rho_2 u_{t-2} + \rho_3 u_{t-3} + \varepsilon_t$ under the alternative.

The BG test checks:

$$H_0 : \rho_1 = \rho_2 = \rho_3 = 0$$

$$H_1 : \text{at least one } \rho_j \neq 0 \text{ (} j = 1, 2, 3 \text{)}$$

Auxiliary regression:

$$\hat{u}_t = \alpha + \delta_1 \text{mon}_t + \delta_2 \text{tues}_t + \delta_3 \text{wed}_t + \delta_4 \text{thurs}_t + \delta_5 t + \delta_6 \text{wave2}_t + \delta_7 \text{wave3}_t + \rho_1 \hat{u}_{t-1} + \rho_2 \hat{u}_{t-2} + \rho_3 \hat{u}_{t-3} + e_t$$

We can write in the following form:

$$\hat{u}_t = \alpha_0 + \gamma' X_t + \rho_1 \hat{u}_{t-1} + \rho_2 \hat{u}_{t-2} + \rho_3 \hat{u}_{t-3} + e_t$$

where $X_t = (\text{mon}_t, \text{tues}_t, \text{wed}_t, \text{thurs}_t, t, \text{wave2}_t, \text{wave3}_t)$

Test Statistic and Distribution:

$$LM = (n - q) \times R_{aux}^2 \sim \chi_{(q)}^2$$

Critical value:

$$\chi_{0.95,3}^2 = 7.815$$

Decision rule: If $LM > \chi_{0.95,3}^2$ or $p < 0.05$, reject H_0 ; Otherwise, fail to reject H_0

Conclusion:

From the R output:

$$LM = 41.127, df = 3$$

Since $41.127 > 7.815$ and $p < 0.05$, reject H_0 . There is strong evidence of third-order autocorrelation in the error term. Hence, the usual OLS standard errors for 2(a) are unreliable; subsequent inference should use autocorrelation-robust (HAC/Newey-West) standard errors or a specification that models the error dynamics.

(d)

We keep the OLS point estimates from 2(a) but replace the usual (homoskedastic, no-autocorrelation) standard errors with **HAC/Newey-West** standard errors allowing **lag = 3**. This is warranted by 2(b)–2(c), which show short-run serial correlation (BG(3) rejects H_0).

The HAC-corrected regression:

$$\begin{aligned} \log(\widehat{\text{avgprc}})_t = & \underset{(0.254)}{-0.954} + \underset{(0.096)}{-0.012 \cdot \text{mon}_t} + \underset{(0.115)}{-0.014 \cdot \text{tues}_t} + \underset{(0.101)}{0.043 \cdot \text{wed}_t} \\ & + \underset{(0.071)}{0.122 \cdot \text{thurs}_t} + \underset{(0.002)}{-0.001 \cdot t} + \underset{(0.024)}{0.095 \cdot \text{wave2}_t} + \underset{(0.020)}{0.049 \cdot \text{wave3}_t} \end{aligned} \quad (4)$$

Comparison with OLS:

The point estimates are identical, since OLS coefficients remain unbiased.

Standard errors are generally larger under HAC correction, reflecting the presence of serial correlation.

wave2 remains strongly significant ($p_{\text{OLS}} = 0.0000$; $p_{\text{HAC}} = 0.0001$).

wave3 remains significant ($p_{OLS} = 0.0220 \rightarrow p_{HAC} = 0.0139$).

The weekday dummy variables and the time trend remain statistically insignificant at the 5% level, though Thursday becomes marginally significant at 10% (HAC $p = 0.0912$).

These adjustments confirm that ignoring serial correlation would understate standard errors and overstate significance.

Conclusion:

After correcting for third-order serial correlation, the overall conclusions remain unchanged: stormy weather significantly raises fish prices, while day-of-week and trend effects are negligible. The HAC/Newey–West correction ensures reliable inference and addresses the autocorrelation detected in Question 2(c).

(e)

Hypothesis

$$H_0 : \beta_{mon} = \beta_{tues} = \beta_{wed} = \beta_{thurs} = 0 \quad H_1 : \text{At least one } \beta_i \neq 0$$

This question examines whether fish prices differ systematically across days of the week. With Friday as the base category, we test the joint null hypothesis. Under the null hypothesis, the test statistic follows an F-distribution with 4 and 89 degrees of freedom:

$$F \sim F(4, 89)$$

At the 5% significance level, the critical value is:

Conclusion:

Individual HAC p-values: Mon 0.9008, Tue 0.9048, Wed 0.6739, Thu 0.0912.

Joint HAC-robust Wald test: typically $F(4, 89) \approx 1.128$ with $p \approx 0.348$

Since $p > 0.05$, fail to reject H_0 . There is no statistically significant evidence that prices vary systematically within a week once we use HAC-robust inference. At most, Thursday shows weak (10%) evidence of a higher price relative to Friday, but the joint test says weekday effects are not collectively significant.

APPENDIX

Question 1(a)

```
hprice1 <- read.csv("hprice1.csv")

model <- lm(price ~ lotsize + sqrft + bdrms + colonial, data = hprice1)

summary(model)
```

Call:

```
lm(formula = price ~ lotsize + sqrft + bdrms + colonial, data = hprice1)
```

Residuals:

Min	1Q	Median	3Q	Max
-122.757	-37.163	-4.639	26.840	228.985

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.166e+01	2.928e+01	-0.740	0.461518
lotsize	2.406e-03	6.744e-04	3.567	0.000603 ***
sqrft	1.213e-01	1.311e-02	9.255	2.02e-14 ***
bdrms	9.590e+00	9.241e+00	1.038	0.302374
colonial	2.007e+01	1.449e+01	1.385	0.169745

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 58.85 on 83 degrees of freedom

Multiple R-squared: 0.677, Adjusted R-squared: 0.6615

F-statistic: 43.5 on 4 and 83 DF, p-value: < 2.2e-16

Question 1(d)

```
# Perform Breusch-Pagan test
library(lmtest)
bp_test <- bptest(model)

# Calculate critical value at 5% significance level
critical_value <- qchisq(0.95, df = bp_test$parameter)
cat("Breusch-Pagan Test Results:\n")
```

Breusch-Pagan Test Results:

```
print(bp_test)
```

studentized Breusch-Pagan test

data: model

BP = 11.36, df = 4, p-value = 0.0228

```
cat("\nCritical value (5% significance level):", critical_value, "\n")
```

Critical value (5% significance level): 9.487729

Question 1(e)

```
m1 <- lm(price ~ lotsize + sqrft + bdrms + colonial, data = hprice1)
```

```
ehat <- residuals(model)
```

```
ehat2 <- ehat^2
```

```
white_aux <- lm(ehat2 ~ lotsize + sqrft + bdrms + colonial +  
                I(lotsize^2) + I(sqrft^2) + I(bdrms^2) +  
                lotsize:sqrft + lotsize:bdrms + lotsize:colonial +  
                sqrft:bdrms + sqrft:colonial + bdrms:colonial,  
                data = hprice1)
```

```
n <- nrow(hprice1)
```

```
r2_white <- summary(white_aux)$r.squared
```

```
LM_white <- n * r2_white
```

```
df_white <- 13
```

```
p_value_white <- 1 - pchisq(LM_white, df_white)
```

```
LM_white;p_value_white
```

[1] 30.78499

[1] 0.003624315

Question 1(f)

```
model <- lm(price ~ lotsize + sqrft + bdrms + colonial, data = hprice1)

summary(model)
```

Call:

```
lm(formula = price ~ lotsize + sqrft + bdrms + colonial, data = hprice1)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-122.757	-37.163	-4.639	26.840	228.985

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.166e+01	2.928e+01	-0.740	0.461518
lotsize	2.406e-03	6.744e-04	3.567	0.000603 ***
sqrft	1.213e-01	1.311e-02	9.255	2.02e-14 ***
bdrms	9.590e+00	9.241e+00	1.038	0.302374
colonial	2.007e+01	1.449e+01	1.385	0.169745

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 58.85 on 83 degrees of freedom

Multiple R-squared: 0.677, Adjusted R-squared: 0.6615

F-statistic: 43.5 on 4 and 83 DF, p-value: < 2.2e-16

```
library(sandwich)
library(lmtest)
robust_se <- coeftest(model, vcov = vcovHC(model, type = "HC1"))

results_comparison <- cbind(
  Coef = coef(model),
  SE_usual = summary(model)$coefficients[, 2],
  SE_robust = robust_se[, 2],
  t_usual = summary(model)$coefficients[, 3],
  t_robust = robust_se[, 3]
)

print(results_comparison)
```

	Coef	SE_usual	SE_robust	t_usual	t_robust
(Intercept)	-21.658225482	2.927639e+01	37.341076203	-0.7397847	-0.5800107
lotsize	0.002405666	6.744394e-04	0.001316637	3.5669128	1.8271298
sqrft	0.121333274	1.310977e-02	0.017256084	9.2551777	7.0313332
bdrms	9.589858128	9.240508e+00	8.833771544	1.0378064	1.0855905
colonial	20.074729875	1.449375e+01	15.937370766	1.3850609	1.2596011

Question 1(g)

```
model2 <- lm(log(price) ~ log(lotsize) + log(sqrft) + bdrms + colonial, data = hprice1)
summary(model2)
```

Call:

```
lm(formula = log(price) ~ log(lotsize) + log(sqrft) + bdrms +
    colonial, data = hprice1)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.67583	-0.10699	-0.00189	0.10014	0.71810

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.24202	0.65379	-1.900	0.0609 .
log(lotsize)	0.17926	0.03870	4.632	1.33e-05 ***
log(sqrft)	0.67827	0.09302	7.292	1.64e-10 ***
bdrms	0.02872	0.02833	1.014	0.3137
colonial	0.05433	0.04428	1.227	0.2232

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1841 on 83 degrees of freedom

Multiple R-squared: 0.6417, Adjusted R-squared: 0.6244

F-statistic: 37.16 on 4 and 83 DF, p-value: < 2.2e-16

Question 1(h)

```
# Estimate the original model from part (g)
model_log <- lm(log(price) ~ log(lotsize) + log(sqrft) + bdrms + colonial, data = hprice1)

# Obtain squared residuals
```

```

residuals_sq <- residuals(model_log)^2

# Run auxiliary regression
aux_reg <- lm(residuals_sq ~ log(lotsize) + log(sqrft) + bdrms + colonial, data = hprice1)

# Calculate LM test statistic
n <- nrow(hprice1)
r_sq_aux <- summary(aux_reg)$r.squared
LM_stat <- n * r_sq_aux

# Critical value at 5% significance level
critical_value <- qchisq(0.95, df = 4)
LM_stat

```

[1] 6.606256

Question 2 (a)

```

# Load data
fish <- read.csv("fish.csv")

# Estimate Q2(a) model
m2a <- lm(log(avgprc) ~ mon + tues + wed + thurs + t + wave2 + wave3, data = fish)

# Usual OLS results
summary(m2a)

```

Call:

```
lm(formula = log(avgprc) ~ mon + tues + wed + thurs + t + wave2 +
    wave3, data = fish)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-1.12979	-0.21917	0.04037	0.22238	0.72876

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.953590	0.189361	-5.036	2.47e-06 ***
mon	-0.012018	0.114115	-0.105	0.916
tues	-0.013734	0.112053	-0.123	0.903

wed	0.042820	0.111712	0.383	0.702
thurs	0.121922	0.110899	1.099	0.275
t	-0.001015	0.001388	-0.731	0.466
wave2	0.094561	0.021779	4.342	3.73e-05 ***
wave3	0.049046	0.021039	2.331	0.022 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3491 on 89 degrees of freedom

Multiple R-squared: 0.3209, Adjusted R-squared: 0.2674

F-statistic: 6.007 on 7 and 89 DF, p-value: 9.552e-06

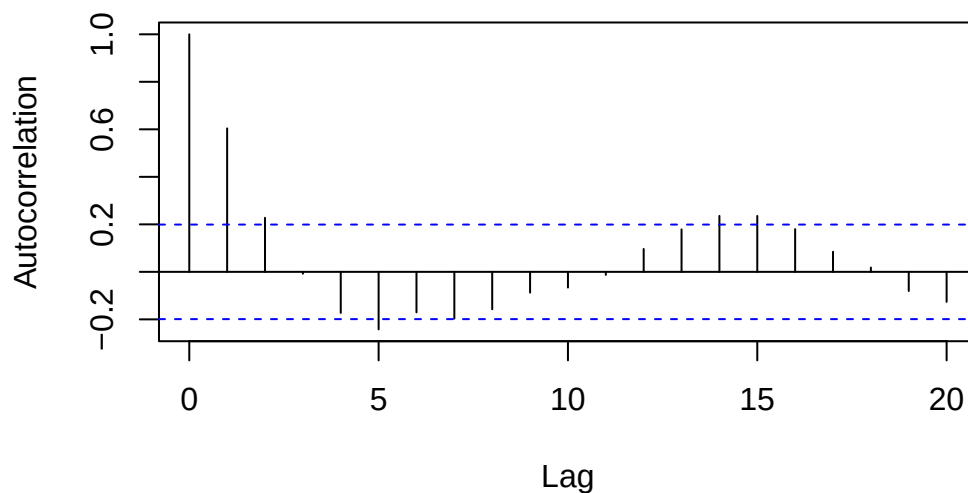
Question 2 (b)

```
# Fit Q2(a) model (as before)
m2a <- lm(log(avgprc) ~ mon + tues + wed + thurs + t + wave2 + wave3, data = fish)

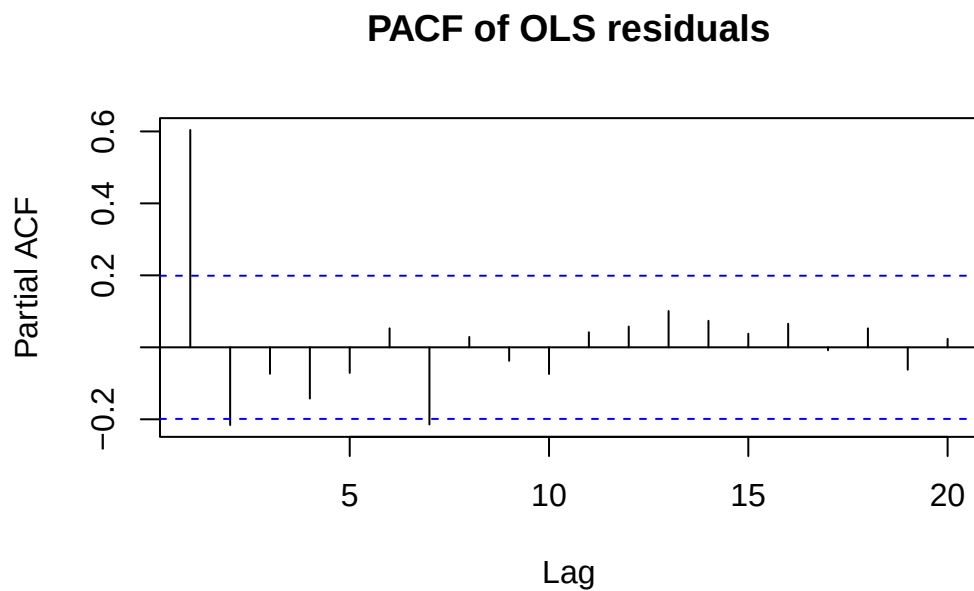
u_hat <- resid(m2a)

# Plot correlogram (ACF)
acf(u_hat, main = "Correlogram of OLS Residuals from Model 2(a)",
    lag.max = 20, ylab = "Autocorrelation", xlab = "Lag")
```

Correlogram of OLS Residuals from Model 2(a)



```
pacf(u_hat, lag.max = 20, main = "PACF of OLS residuals")
```



Question 2 (c)

Breusch-Godfrey test for serial correlation of order up to 3

data: m2a

LM test = 41.127, df = 3, p-value = 6.147e-09

Breusch-Godfrey test for serial correlation of order up to 3

data: m2a

LM test = 21.101, df1 = 3, df2 = 86, p-value = 2.465e-10

[1] 7.814728

Question 2 d

Table 2: Comparison of OLS and HAC (Newey–West, lag = 3) Results

term	estimate	se_OLS	t_OLS	p_OLS	se_HAC	t_HAC	p_HAC	sig_HAC
(Intercept)	-0.9536	0.1894	-5.0358	0.0000	0.2544	-3.7487	0.0003	***
mon	-0.0120	0.1141	-0.1053	0.9164	0.0961	-0.1251	0.9008	
tues	-0.0137	0.1121	-0.1226	0.9027	0.1146	-0.1199	0.9048	
wed	0.0428	0.1117	0.3833	0.7024	0.1014	0.4222	0.6739	
thurs	0.1219	0.1109	1.0994	0.2746	0.0714	1.7074	0.0912	.
t	-0.0010	0.0014	-0.7314	0.4664	0.0020	-0.5137	0.6088	
wave2	0.0946	0.0218	4.3419	0.0000	0.0235	4.0254	0.0001	***
wave3	0.0490	0.0210	2.3312	0.0220	0.0195	2.5099	0.0139	*

Question 2 (e)

Linear hypothesis test:

mon = 0

tues = 0

wed = 0

thurs = 0

Model 1: restricted model

Model 2: $\log(\text{avgprc}) \sim \text{mon} + \text{tues} + \text{wed} + \text{thurs} + \text{t} + \text{wave2} + \text{wave3}$

Note: Coefficient covariance matrix supplied.

	Res.Df	Df	F	Pr(>F)
1	93			
2	89	4	1.1282	0.3484